

AI for Cybersecurity

SIL765

Internet-of-Things (IoT)

Smart City



Smart Home



Smart Transportation



Smart Robot



Cyber Attacks

Ukraine power cut 'was cyber-attack'¹

Mirai botnet: How CCTV cameras almost brought down the internet²



Hackers remotely kill a Jeep on the highway³

'I'm in your baby's room':
A hacker took over a baby monitor⁴

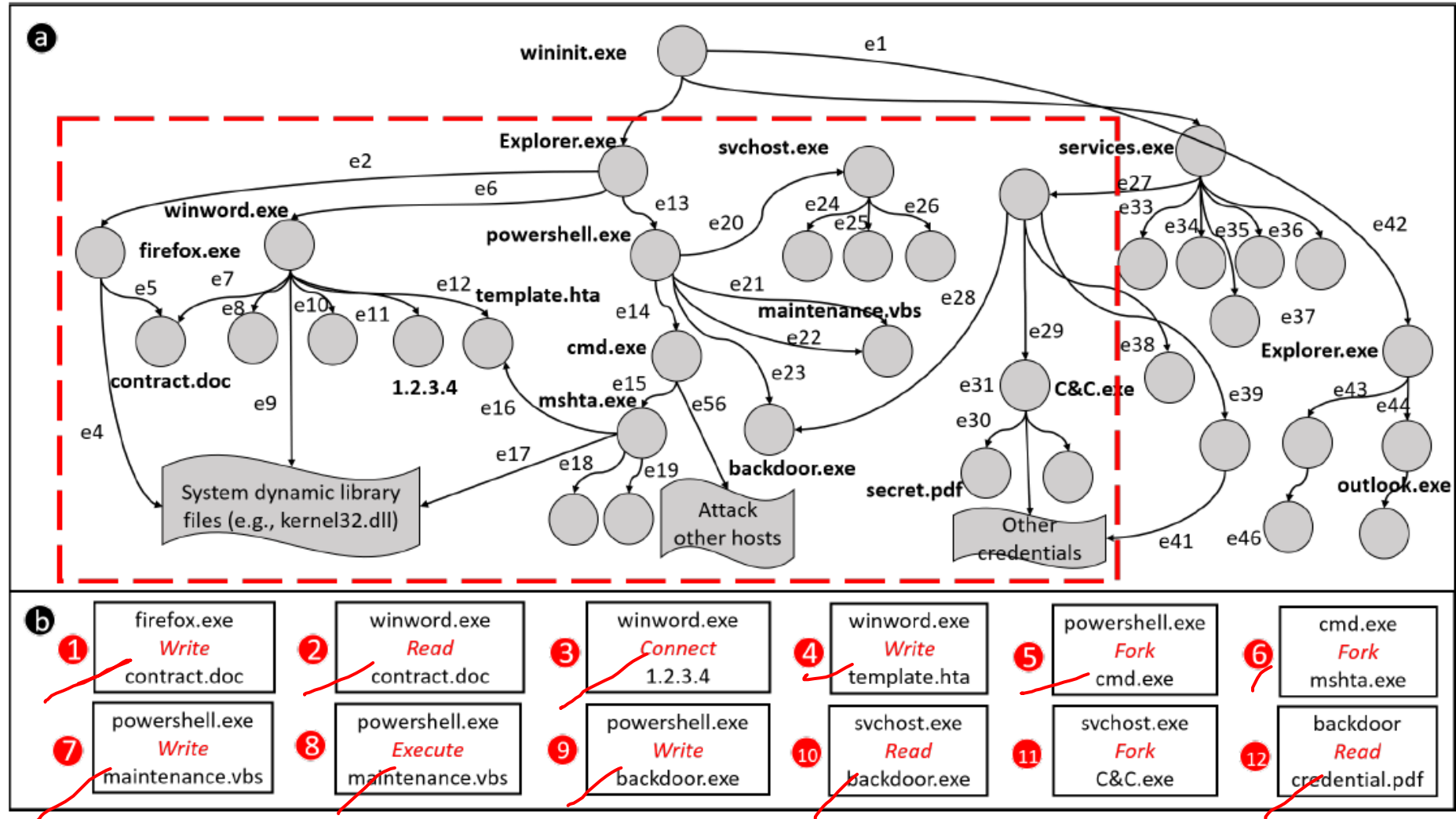
Bugs

- Uninitialized variables
- Null pointer dereference
- Use after free
- Double free
- Array indexing errors
- Mismatched array new/delete
- Potential stack overrun
- Potential heap overrun
- Return pointers to local variables
- Logically inconsistent code
- Invalid use of negative values
- Passing large parameters by value
- Under-allocations of dynamic data
- Memory leaks
- File handle leaks
- Network resource leaks
- Unused values
- Unhandled return codes
- Use of invalid iterators

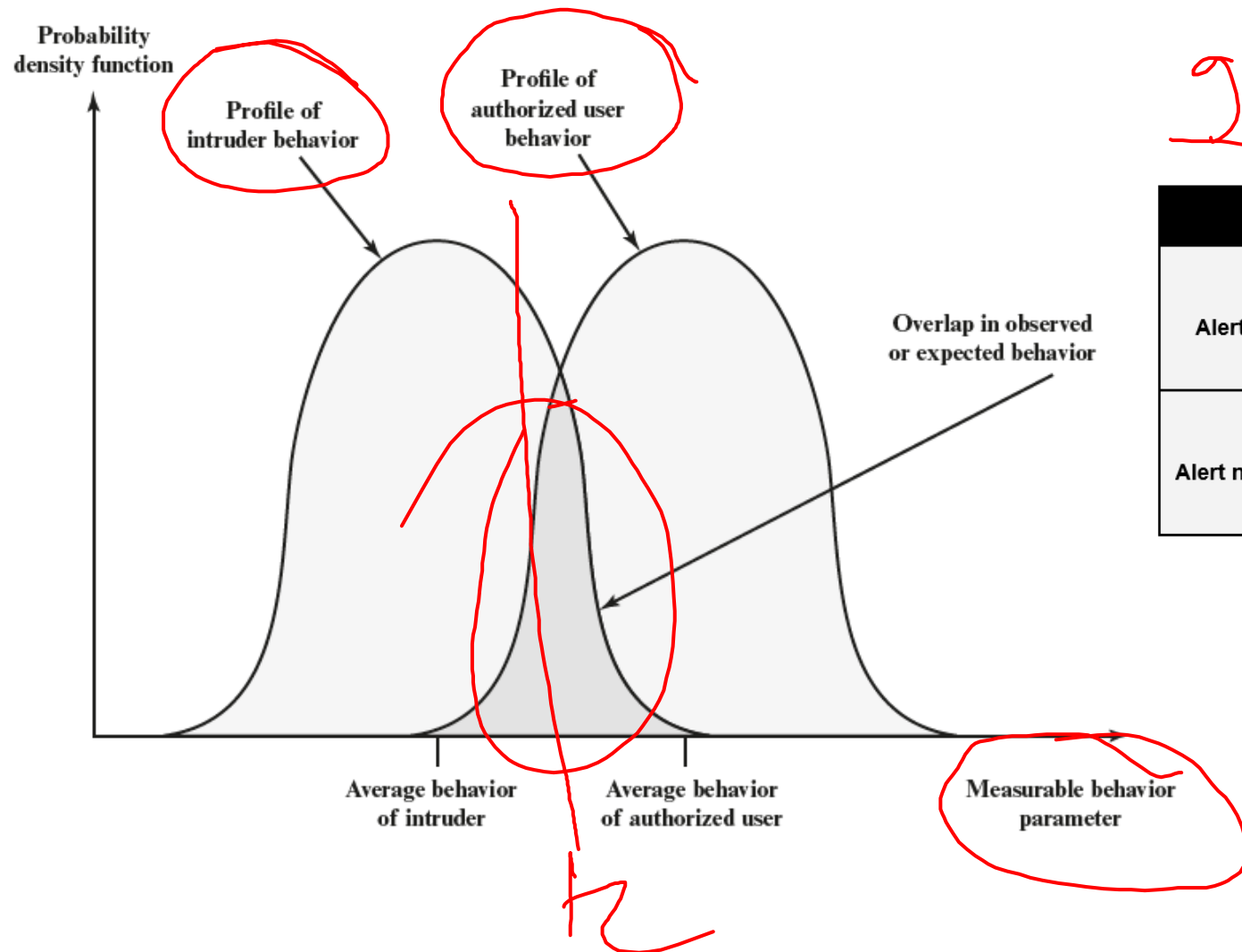
Intrusion Detection System (IDS)

- Signature-based
- Conventional statistics-based classification
- Conventional anomaly-based detection
- Modern ML/AI-based classification/anomaly-based detection

Signature-Based IDS



Classification-Based IDS



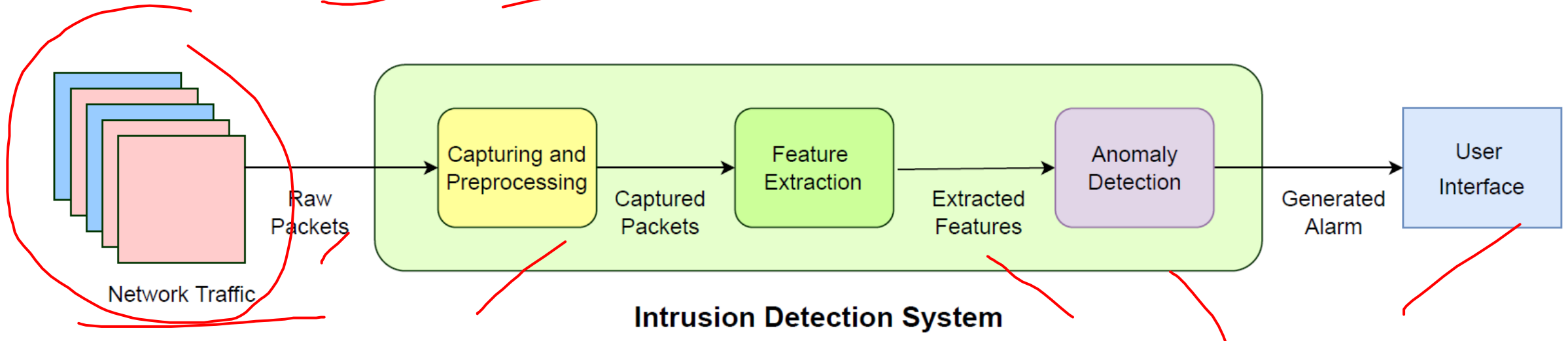
	Occurrence of malicious activity	No occurrence of malicious activity
Alert generated	True Positive Occurrence of malicious activity identified by the monitoring system	False Positive No malicious activity, but identified by the monitoring system
Alert not generated	False Negative Occurrence of malicious activity not identified by the monitoring system	True Negative No malicious activity and not identified by the monitoring system

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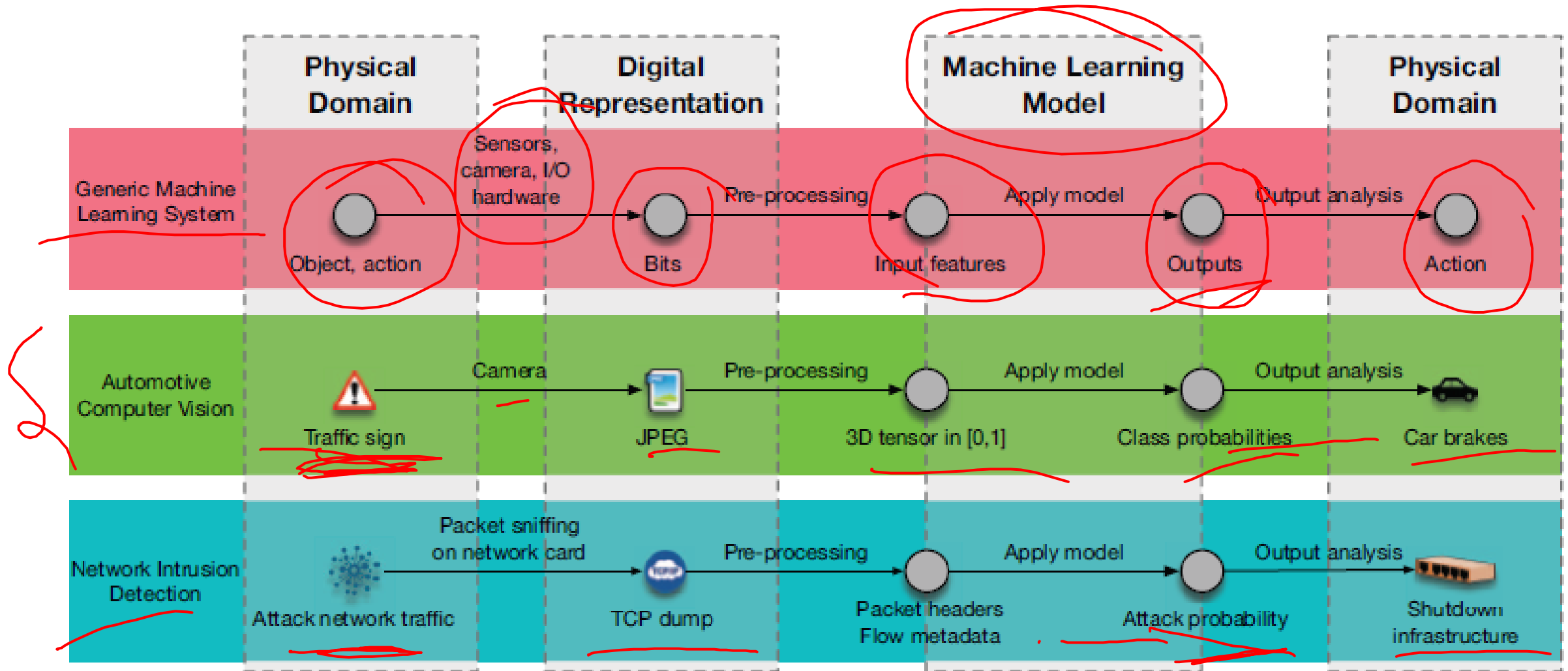
Anomaly-Based IDS

- Only the distribution of the benign behavior is known

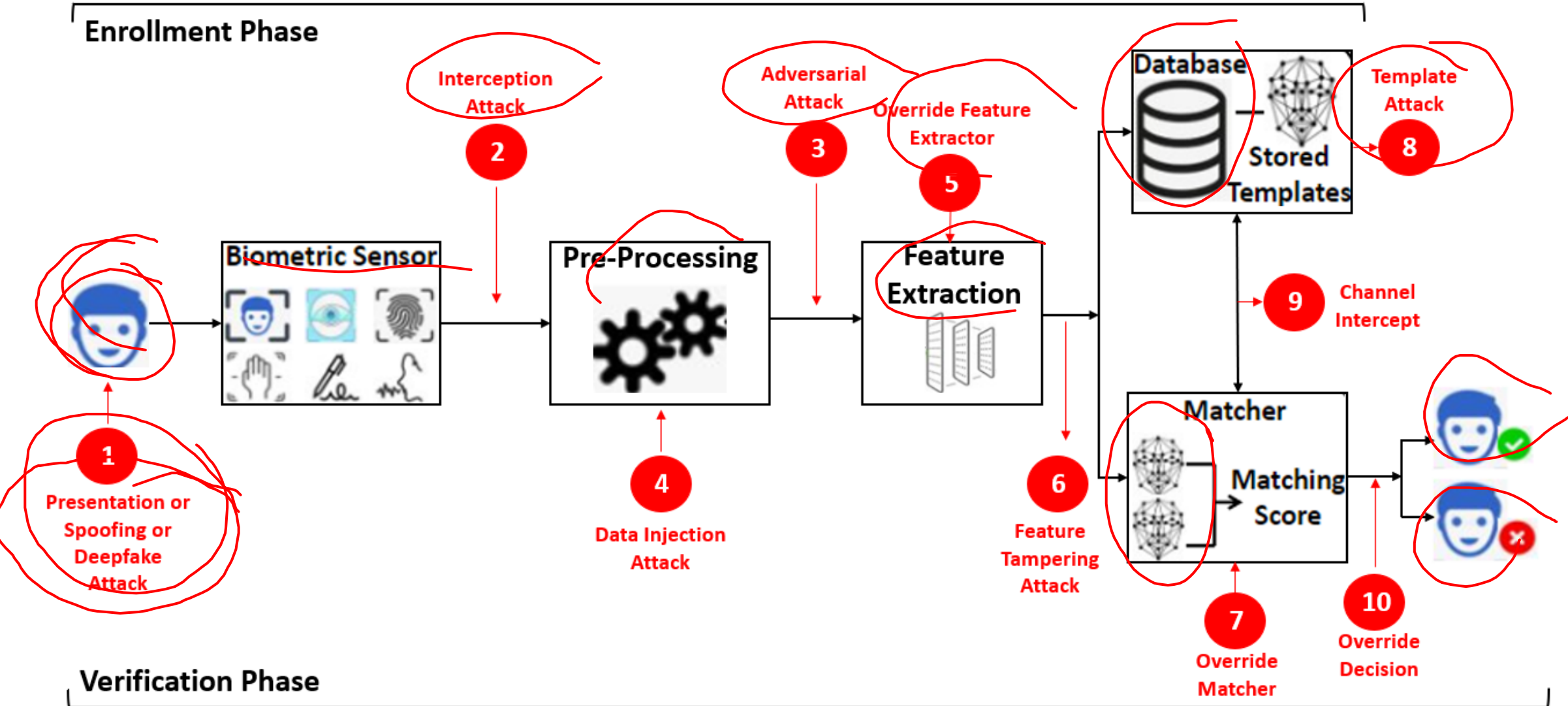
INIIDS, SIIDS



ML/AI-based IDS



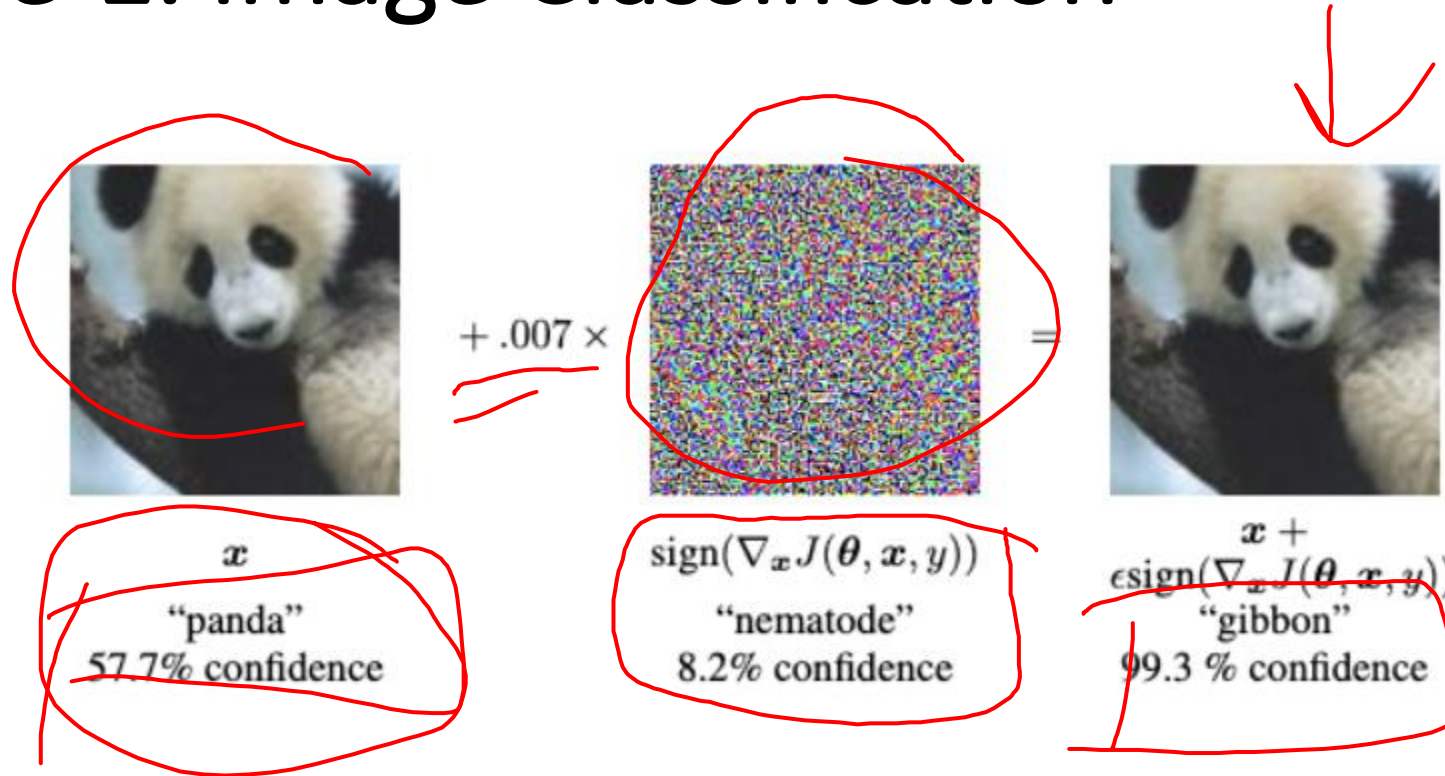
Attacks Points in a Typical ML System



Adversarial ML: Threat Model

- Goal
 - Affect output in the test phase
- Capability
 - Test phase input
 - Test phase input and train phase input (**training set poisoning**)
- Knowledge
 - Test phase output (**Blackbox**)
 - Test phase output, train phase input and output (Blackbox)
 - Test phase output distribution, train phase input and output (Greybox)
 - Model weights (Whitebox)

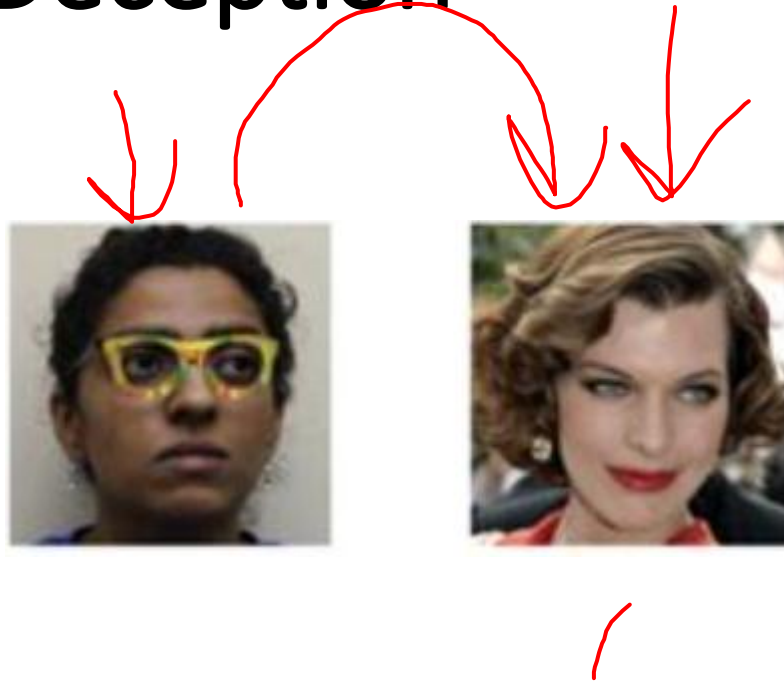
Example-1: Image Classification



Example-2: Image Detection



Example-3: Deception

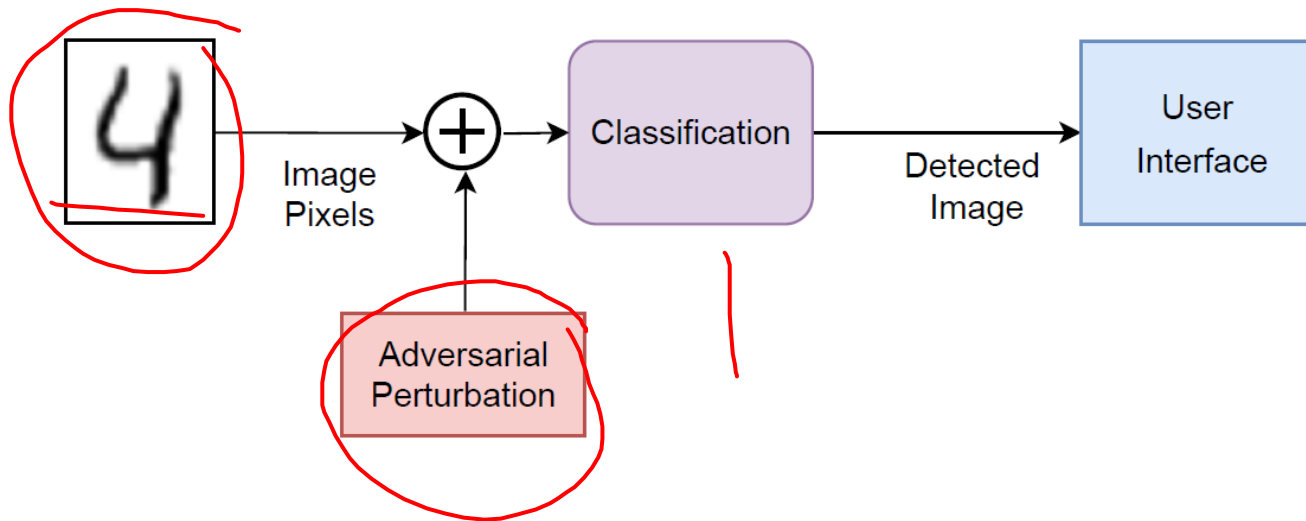


Example-4: Speech Recognition

- <https://mosis.eecs.utk.edu/advpulse.html>

AML Summary

- Make carefully crafted perturbations in the image pixels
- Make ML model output a wrong class during the inference phase
- AML-assisted attacks have been generalized to different domains



AML Attacks against IDS

- Goal: Evade IDS during the inference phase
- Methodology: Make carefully crafted perturbations in the network packets
- Outcome: AML-assisted attacks can be a severe threat to the IoT

